Physico-chemistry of solutions in pharmacy

I) Study of solutions:

1) Definition:

⇒ Solution is a homogeneous mixture, composed of two or more substances

⇒ It consists of:

⇒ **Solute** is the substance being dissolved

⇒ **Solvent** is the substance doing the dissolving

* The solute is typically present in a smaller amount compared to the solvent

2) Characteristics of a solution:

⇒ **Homogeneity**: solutions are uniform throughout, meaning that the solute particles are evenly distributed within the solvent

⇒ **Particle size**: The particles in a solution are at molecular or ionic level, usually smaller than 1nm, which prevents them from being visible to the naked eye

⇒ **Stability**: solutions are generally stable; the solute does not settle out (sediment) over time, nor can it be easily separated by filtration

⇒ **Transparency**: Most solutions are clear and transparent, though the color may vary depending on the solute

3) Types of solutions:

⇒ **Aqueous solutions**: water = solvent

⇒ **Non-aqueous solutions**: water ≠ solvent such as alcohol or ether

⇒ **Gas solutions**: such as air, which is a solution of gases

⇒ **Solide solutions**: like alloys, where metals are dissolved into each other in the solide phase

4) Different modes of expressing concentration:

1) **mass concentration**: $C = m/V$ (g/l)

The mass of solute per unit volume of solution

m: mass of solute (g) — V: Volume of solution (l)

2) **Molar concentration: (Molarity)**

The number of moles of solute per liter of solution

$C_M = n/V$ (mol/l or M)

$n = m/M$, $C_M = m/M.V$

$C_M = \frac{10 \times d}{M}$

3) **Molal concentration: (Molality)**

the number of moles of solute per Kilogram of solvent (mol/kg)

$C_{molale} = \frac{n_{solute}}{m_{solvant}\ (kg)}$

4) **Parts per Million: (ppM)**

The number of parts of solute per one million parts of solution, often used for very dilute solutions

$C_{PPm} = \frac{m_{solute}}{m_{solution}} \times 10^6$

$= \frac{\sim}{\sim} \times 10^6 = \ldots$

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5) Percentage concentrations:

⇒ Weight / Weight Percentage (W/W):

$$(W/W)\% = \frac{m_{solute}}{m_{solution}} \times 100$$

⇒ Weight / volume Percentage (W/V):

$$(W/V)\% = \frac{m_{solute}}{V_{solution}} \times 100$$

(commonly used for solid solutes in liquid)

⇒ Volume / Volume Percentage (V/V):

$$(V/V)\% = \frac{V_{solute}}{V_{solution}} \times 100$$

5) Density: $\rho = m/V$

The mass of substance pa unit volume, it is typically in units such (g/cm^3) or (kg/m^3)

The mole fraction: (الكسر المولي)

n_1: nombre de mole de soluté

n_0: " " de solvat

$f_1 = \frac{n_1}{n_1+n_0}$ = fraction of solute

$f_2 = \frac{n_0}{n_1+n_0}$ = fraction of solvat

$f_1 + f_2 = 1$

التمديد: $C_1 = C \times \frac{\text{شحال دنا}}{\text{حنيق}}$

II) Electrolytic solutions: 1) Definition:

are solutions that contain ions and can conduct electricity due to the presence of dissolved electrolytes (is a substance that when dissolved in water or another solvant, dissociates into positively charged ions (cations) and negatively charged ions (anions)

2) characteristics of electrolytic solutions:

a) Ionization: Electrolytes dissociate into ions when they dissolve in a solvent, allowing the solution to carry an electric current

b) conductivity: electrolytic solutions can conduct electricity because the mobile ions can move freely within the solution, enabling the flow of electric charge

3) Types of electrolytes:

a) strong electrolytes: These completely dissociate into ions in solution ($\alpha = 1$)

b) weak electrolytes: These only partially dissociate into ions ($\alpha = 0$)

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it of concentration is g/l
10^{-3} → Kg

3) Concentration in equivalents per liter:
is a measure of the quantity of dissolved solute that takes into account the ion's capacity to react or participate in chemical reactions. In chemistry, equivalents are used to express ion concentration based on their ability to donate or accept electrons (in redox reactions) or hydrogen ions (in acid-base reactions)

$$C_{Eq/l} = \frac{n_{Solute} \times Valence}{V_{solution}} \rightarrow \text{ions charge or oxidation state}$$

$$= C_M \times Valence$$

4) Normality (N):
The number of equivalents of solute per liter of solution, often used in acid-base reactions

$$N = M \times \text{Equivalent factor}$$

refers to the number of (H^+) or (OH^-) released or accepted in an acid-base reaction, or the number of electrons exchanged in a redox reaction

* Difference between Normality and molarity: N and M are both measures of solution concentration (specific chemical reaction and the n.f of the solute) the relationship between N and M depends on the

M: is the number of moles of solute per 1l of solution

N: is the number of equivalents of solute per 1l of

4) Dissociation coefficient:
→ α is the ratio of the concentration of dissociated species (ions) to the initial concentration of the solute before dissociation

$$\alpha = \frac{C_1 \text{ (ions)}}{C_0 \text{ (solute)}}$$

→ The dissociation coefficient ranges from 0 to 1
α = 0 (no dissociation → none of the solute has dissociated)
α = 1 (complete dissociation → all of the solute has dissociated into ions)
→ The value of α can depend on various factor (temperature, pressure, the nature of the solute and solvent)
→ α is important in understanding the behavior of electrolytes in solution, as it helps predict the solution's conductivity, colligative properties and chemical reactivity

5) Different modes of expressing concentration:
1) osmolar concentration (osmolarity):

osmol/l $\left(C_{osmolar} = \frac{\Sigma n \text{ (at equilibrium)}}{V_{solution}} \right.$

* the total concentration of all solute particles of solution (number of osmoles per liter of solution), expressed in osmoles per liter

2) Osmolal concentration (osmolality):

osmol/kg $\left(C_{osmolale} = \frac{\Sigma n}{m_{solvent\,(kg)}} \right.$

Number of moles of all solute particles per kg of solvent

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$\ell \xrightarrow{10^{-3}} m^3$

So: the unit of concentration in the International System is mol/m³

5) Ionic Strength (μ):

Defined for electrolyte solution as:

$$\mu = \frac{1}{2} \sum C_i \cdot Z_i^2$$

C_i: ions concentration

Z_i: the valance

M_c

m_{Sc}

m_{solu}

m_{soluti}

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